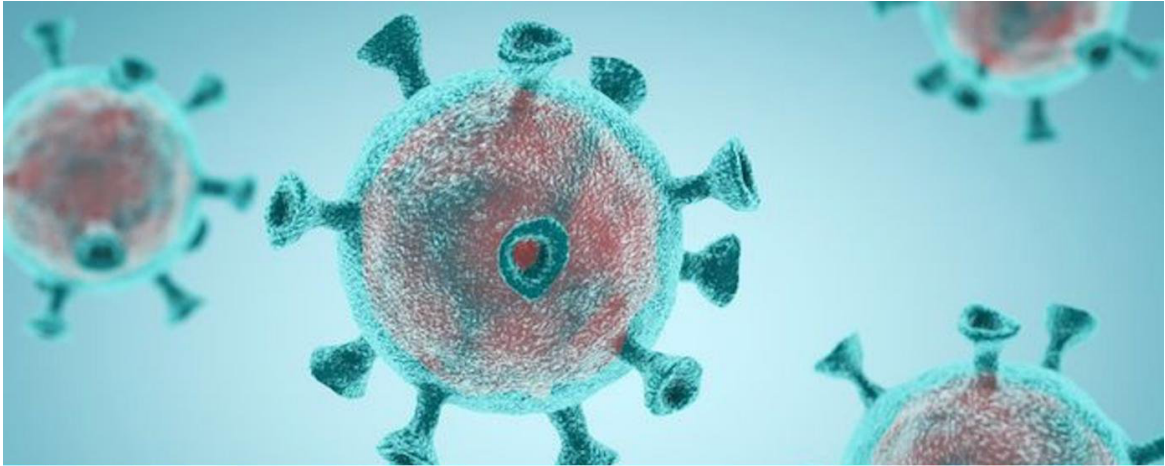


Machine Learning for Design



Assignment 2 Group:

G 4

Members: Vlad Dobriakov, Josha Leemans, Karlijn Tangelder, Jop van der Post & Bing Zwikker

Introduction

During the covid pandemic the shift to remote learning impacted the student experience at the TU Delft. In this report we are challenged with improving this covid-era experience, our group used text processing and machine learning techniques to analyse the anonymous student feedback. Our analysis is ranging from VADER sentiment analysis to topic modelling and revealed a returning pattern of feeling of isolation. Students frequently highlighted "online lectures," "working from home," and a significant "lack of social contact" as their primary obstacles during this covid-era, leading to a measurable decline in mental well-being and academic drive.

Solving this problem is critical, since the pandemic did more than just transfer the lectures to online. It took away the social aspect of learning. Our sentiment analysis showed a negative trend, where even seemingly positive comments often asked "needs for support". The lack of motivation and peer connection doesn't just hurt grades, it also leads to a disconnected campus culture where the collaborative aspect is lost.

To end this gap we propose the TUD Social Study Hub. This solution consists of a digital matching platform that pairs students based on courses and interests to work together. Also physical study spaces, this are zones on campus specially designed for students to create a collaborative growth. And lastly autonomous networking, this is a feature that allows students to build their own academic social circles.

During the covid pandemic students couldn't come to the faculty, which meant they had to study from their home. The students at the TU Delft for example were asked to help with a project to improve the bachelors students' experience during the covid-era. Students had to share their experience anonymously and tell what the main problem was they encountered. With text processing skill we as a group analysed the text to find ways to improve the students' experience in these difficult times.

Once our solutions are used on campus, we expect the student experience to shift from isolation to collaborative growth. The success of this solution will be seen back in future data, where our models will see clusters like: "community", "support" and "together". Ultimately students won't be sitting behind their desk all day, but will actually feel part of a connected campus, whether they are studying at home or at TU Delft.

Text analysis

Text preprocessing

To ensure the analysis is reliable, a preprocessing pipeline was used. The pre-processing stage is very important, as machine learning models require structured and numerical input: raw text, however, is unstructured and cannot be processed directly. This is because the text data cannot be processed by ML models in one go, as it consists of unstructured text rather than numerical features. The aim of preprocessing is therefore to reduce noise and convert the text into more comparable and meaningful tokens. The preprocessing pipeline consists of the following steps:

- 1) **Lowercasing:** all text has been converted to lower case to ensure consistency, so that "Home" and "home" are treated as the same word. This results in less redundancy and reduces the dimensionality of the dataset.
- 2) **Tokenisation:** the sentences have been split into individual words, or tokens, allowing the text to be analysed at word level, which is very important for topic and frequency analysis.
- 3) **Removal of punctuation and numbers:** numbers and punctuation have been removed, as they do not contribute to meaningful semantic information and also because they introduce noise into the analysis.
- 4) **Stopword removal:** standard English stopwords, such as "and", "the" and "is", have been removed using a predefined stopwords list from the NLP library. These words occur frequently but convey little meaning, which can distort the analysis.
- 5) **Custom stopwords:** additional domain-specific words have also been removed, such as "TU", "student", "study" and "Delft", as these words occur very frequently in the dataset but do not contribute to distinguishing between different topics, thereby altering the results.
- 6) **Stemming and lemmatisation:** the words have been reduced to their base form, such as "studying" to "study", through lemmatisation and stemming. This reduces the variation in word forms and does the same for sparsity in the dataset used. This results in a more robust analysis, whilst it may also reduce nuance.

Implementation

The preprocessing was executed using the following function:

```
tokens = [  
    preprocess(sentence, lower=True,  
    word_tokenization=True,  
    rem_punc=True, rem_numb=True,  
    rem_stopwords=True,  
    extra_stopwords=["people",  
    "go", "becaus", "n't", "realli", "student", "tu", "delft", "university", "study", "studies", "day", "year", "time", "we  
    ek", "period", "veri", "peopl", "studi", "lot", "littl", "feel", "problem", "difficult", "thing", "teacher"],  
    stem=True,  
    lem=True) for sentence in corpus  
]  
  
print(tokens)
```

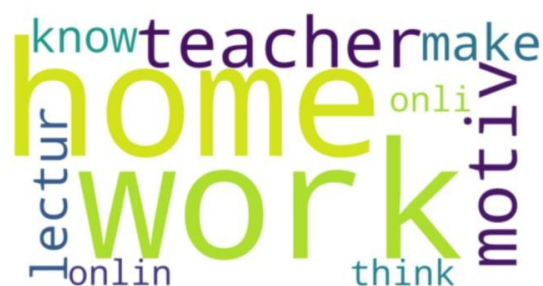
The function described above is based on the implementation from the course tutorial and utilises standard NLP techniques, such as tokenisation, stop-word removal, lemmatisation and stemming.

Reflection on pre-processing

The pre-processing has significantly improved the dataset by standardising words and removing noise. One potential limitation is that the combination of stemming and lemmatisation can result in a reduction of semantic nuances. However, this trade-off is accepted, as the result leads to clearer recognition of patterns in the data.

Word Frequencies Analysis

The first exploratory step involved a word frequency analysis, which was carried out to identify the most frequently occurring words in the dataset.



Following preprocessing, words such as “work”, “home”, “teacher”, “lecture” and “motivation” are more clearly visible. Initially, the most frequent words consisted mainly of stop words. Using these insights, the pre-processing pipeline was fine-tuned by expanding the list of stop words and applying lemmatisation and stemming. This adjustment improved the quality of the results and led to a more meaningful distribution of the words.

Interpretation: these words show that students mainly talk about the following at home:

- Studying from home
- Online learning
- Problems with motivation

In the context of COVID-19, this indicates that home study and online learning are key topics.

Limitation: the word frequency analysis treats these words as independent of one another and therefore does not take into account the context or relationships between the words. As a result, this method provides only a simplified representation of the actual data.

Sentiment Analysis

After doing the word frequency analysis we started the sentiment analysis, in which we used VADER to analyse the amount of negative, positive and neutral sentiments in the provided dataset.

VADER was chosen because this model has been specifically designed for shorter, more informal texts and also because it produces good results with subjective data, such as student feedback. Furthermore, it generates interpretable scores, such as negative, neutral and positive scores as well as a composite score.

VADER assigns sentiment scores to each text based on a predefined lexicon of words that are associated with positive or negative emotions. Each sentence receives a positive, neutral, and negative score, as well as a compound score that represents the overall sentiment of the statement. The breakdown of the sentiment scores is as follows: negative (280), neutral (108) and positive (233). From this analysis we found that the statements can mainly be divided into two groups. Most of the responses contain a negative sentiment, which indicates that many students describe problems or difficulties. However, there are also a considerable number of positive statements, while neutral statements occur less frequently.

When examining the positive statements more closely, it becomes clear that some of them contain a double meaning. For example, the sentence “would love TU Delft could help find good psychologist” receives a positive sentiment score because of the phrase “would love”. However, the actual meaning of the sentence is more neutral or even slightly negative, because it expresses a need for psychological support. This illustrates the concept of semantic ambiguity, where certain words can carry a positive sentiment in the lexicon but represent a different meaning in the context of the sentence. This shows that VADER provides a useful general overview of sentiment, although it can misinterpret context-dependent meanings. Even with this limitation, the sentiment analysis clearly shows that negative experiences dominate the dataset.

```
sentiment
negative      280
positive      233
neutral       108
Name: count, dtype: int64
```

Frequency of sentiment analysis being positive, negative or neutral.

		sentences	neg	neu	pos	\
526	realli like questionair hope help	0.0	0.198	0.802		
254	would like free coffe campus	0.0	0.341	0.659		
161	would interest result well check	0.0	0.370	0.630		
584	better workplac cours home veri nice	0.0	0.412	0.588		
580	would love tu delft could help find good psych...	0.0	0.419	0.581		
	compound sentiment					
526	0.7964	positive				
254	0.7003	positive				
161	0.6249	positive				
584	0.6908	positive				
580	0.9100	positive				

Table with most positive sentiment analysis.

		sentences	neg	neu	pos	\
598	loneli panic attack hopeless frustrat	0.825	0.175	0.0		
534	anxieti stress lack motiv concentr problem fatigu	0.661	0.339	0.0		
607	loneli boredom stress fear see famili angri en...	0.655	0.345	0.0		
355	concentr problem lack social interact poor qua...	0.618	0.382	0.0		
590	veri sick corona veri tire aros psychic proble...	0.605	0.395	0.0		
	compound sentiment					
598	-0.8555	negative				
534	-0.7783	negative				
607	-0.8860	negative				
355	-0.7964	negative				
590	-0.8481	negative				

Table with most negative sentiment analysis.

		sentences	neg	neu	pos	\
11	wait time tu delft psychologist decreas	0.0	1.0	0.0		
441	daytim activ outsid door bond univers	0.0	1.0	0.0		
445	communic especi longer period onli digit reall...	0.0	1.0	0.0		
444	littl motiv studi home lot	0.0	1.0	0.0		
436	tuition fee reduc durat corona crisi	0.0	1.0	0.0		
	compound sentiment					
11	0.0	neutral				
441	0.0	neutral				
445	0.0	neutral				
444	0.0	neutral				
436	0.0	neutral				

Table with most neutral sentiment analysis.

TOPIC MODELING

```
[Topic 0]: 0.011*"also" + 0.011*"work" + 0.009*"would" + 0.008*"onlin" + 0.007*"get" + 0.007*"think" + 0.007*"could"
[Topic 1]: 0.016*"home" + 0.014*"contact" + 0.013*"work" + 0.011*"less" + 0.010*"lectur" + 0.010*"physic" + 0.009*"get"
[Topic 2]: 0.012*"would" + 0.011*"like" + 0.010*"onli" + 0.008*"work" + 0.008*"make" + 0.008*"think" + 0.007*"also"
[Topic 3]: 0.017*"also" + 0.014*"work" + 0.011*"onlin" + 0.010*"contact" + 0.008*"make" + 0.008*"group" + 0.008*"home"
[Topic 4]: 0.010*"also" + 0.010*"onlin" + 0.009*"get" + 0.009*"exam" + 0.009*"much" + 0.008*"place" + 0.008*"home"
[Topic 5]: 0.014*"contact" + 0.012*"fellow" + 0.010*"know" + 0.009*"also" + 0.008*"stress" + 0.007*"ask" + 0.007*"home"
```

Topic modeling can be seen as a more advanced form of word frequency analysis. Instead of only counting how often individual words appear, topic modeling analyzes how often words occur together within the same responses. By analyzing these word co-occurrences, the model groups frequently related words into topics that represent common themes in the dataset. The decision to use topic modelling was made because, unlike word frequency analysis, it can identify relationships between words and makes it possible to identify underlying themes in the data.

After running the model, we identified three topics. Each topic consists of a set of words that frequently appear together in the student responses. The three topics were chosen because this provided a clear and easily explainable distinction between themes, whilst avoiding excessive overlap between different topics.

```
[Topic 0]: 0.012*"would" + 0.009*"also" + 0.009*"get" + 0.008*"work" + 0.008*"make" +
0.008*"lectur" + 0.007*"contact" + 0.006*"fellow" + 0.006*"think" + 0.006*"onlin"

[Topic 1]: 0.011*"also" + 0.009*"work" + 0.009*"onlin" + 0.009*"home" + 0.007*"make" +
0.007*"much" + 0.007*"get" + 0.007*"think" + 0.007*"like" + 0.006*"exam"

[Topic 2]: 0.014*"contact" + 0.011*"work" + 0.011*"also" + 0.009*"home" + 0.007*"much" +
0.007*"onlin" + 0.007*"fellow" + 0.007*"social" + 0.007*"like" + 0.006*"less"
```

When interpreting these topics, several recurring themes become visible. Many of the words are related to working online and studying from home, such as online, home, and lecture. In addition, words such as contact, fellow, and social appear frequently, which suggests that students often mention the lack of social interaction with other students.

Overall, these topics indicate that the main themes in the dataset are related to online education from home and the lack of social contact between students.

These findings are consistent with the results from the word frequency analysis (e.g., "home", "online", "work") and the sentiment analysis, where negative emotions dominate. This consistency across multiple methods increases the reliability of the results. By combining word frequency analysis, sentiment analysis and topic modelling, we ensured that both the surface-level patterns and the underlying relationships within the data were captured.

Students' Input

From the text analysis, we identified three important student needs that frequently appear in the dataset. The first important need is more social contact between students. In the topic modelling results, words such as contact, fellow, and social frequently appeared together. Based on this, we conclude that students often indicate that they experience a lack of social interaction with fellow students and that this is perceived as unpleasant. This is also visible in several responses from the dataset. Some students mention that they struggle with the lack of social contact during their studies. An example of this is: *"The lack of social contacts, I came to live new in Delft myself and I have not been able to meet new people yet."*

A second important need mentioned by students relates to online education and studying from home. In the word frequency analysis, words such as lecture, home, work, and online appeared frequently. Many students describe that it is difficult to actively participate in online lectures and indicate that it is less effective than attending lectures physically on campus. The following quote from a student illustrates dissatisfaction with online education: *"Online education is jerk. people don't see is jerk. Sitting behind my monitor 10 hours a day is jerk."*

The third recurring theme is stress and lack of motivation. This mainly appears in the sentiment analysis, where a large portion of the responses received a negative sentiment score. Many students describe feelings such as stress, fatigue, or a lack of motivation. Deadlines and exams are often mentioned as sources of pressure. In addition, many students state that spending long hours behind a screen reduces their motivation. One student expressed this as follows: *"I don't know why it is but I am less motivated since the corona crisis."*

These three needs were selected because they appeared repeatedly across different analysis methods. Both the word frequency analysis and the topic modelling show that themes such as studying from home, online education, and social contact are central in the dataset. In addition, the sentiment analysis confirms that many students share negative experiences related to these themes.

Solution description

Based on the students' responses and our analysis of them, we are proposing a solution to TU Delft in the form of a Social Study Hub. This solution combines a digital platform with physical study spaces that encourage students to study together and, in doing so, meet new people. This solution aims to address three key needs identified through the text analysis: a lack of social interaction among students, difficulties with online learning, and a significant drop in motivation during the COVID-19 pandemic. These needs emerged from several analytical methods: the word frequency analysis revealed terms such as 'home' and 'online'; the topic modelling analysis, on the other hand, identified clusters centred on 'contact' and 'social'; and the sentiment analysis concluded that negative experiences predominated.

The TUD Social Study Hub consists of three components. The first component is a physical study environment on campus where students can study together in small groups. This environment offers spaces for group work, quiet areas, and places where students can attend online lectures together. This allows students to participate in online education without having to do so in isolation at home. The physical study environment serves as a meeting place for students who have been matched through the platform, but also as an open study space where students can meet spontaneously. A clear separation ensures the hub remains accessible and functional for all students, while the digital platform encourages social interaction. Studying together in a physical environment can contribute to more structure in students' lives and greater social interaction. Studying together in a physical environment can help bring more structure to students' lives and foster greater social interaction. This section therefore directly addresses the lack of social interaction identified in the topic modelling results. Furthermore, words such as "social" and "contact" frequently appeared together.

The second component is an online platform where students sign up to study together. Here, students can indicate, for example, when they want to study, which course they are taking, and whether they prefer to work in a group or individually. Using this information, the system can automatically match students who are taking similar courses or want to study at the same time. This lowers the barrier to forming study groups, where students can potentially make new friends. This design choice is based on the difficulty students expressed in making connections whilst undertaking distance learning, as this is reflected both in the negative sentiment scores and the recurring mentions of isolation in the data.

The third component is a matching feature that allows students to discover and select study partners themselves, similar to the matchmaking concepts that are very popular these days. Students can view profiles of other students with similar (academic) interests and decide for themselves with whom they want to study or connect. This component gives students a sense of autonomy in building their social and academic network, in contrast to component 2, where the system holds the autonomy. This can be particularly valuable for students who are new to Delft and have not yet established many social connections. The preferences of students identified in the data address this variation in this section, where some students have a greater need for structured support, whilst others, conversely, place greater value on greater autonomy.

These findings align directly with the results of the text analysis. In the topic modelling, words such as "contact", "fellow" and "social" frequently appeared together, indicating a need for more social interaction. This was also reflected in student responses, such as: "The lack of social contacts, I'm new to Delft myself and haven't been able to meet new people yet." Words such as "online", "home", and "lecture" also appeared regularly in the analysis, suggesting that many students are struggling with prolonged online study from home. By combining physical meeting places with digital

matching, the TUD Social Study Hub can alleviate these issues and give students a nudge to work on their social study environment.

Finally, the success of these components can be measured by reanalysing future surveys using the same text processing techniques. If the solution is effective, we expect words such as “lonely”, “stress”, or “lack of contact” to appear less frequently in students’ responses, while words such as “support”, “community”, “together” or “study group” will appear more often. This would indicate that student well-being has improved thanks to the TUD Social Study Hub.

Conclusion

In this report, we looked at what students needed during COVID. We did this by using computer tools to process text. First, we cleaned up the text, then we counted how often certain words appeared. We also checked how people felt based on their words and figured out the main topics they were talking about. It turns out students are mostly having trouble with not getting enough social time, finding online learning tough, and feeling less motivated and generally not as good. So, with all this in mind, we came up with the TUD Social Study Hub. It's basically a mix of real, physical places to study and an online tool that helps people find others to work with, fostering teamwork and social ties. It covers the main things, but it's not perfect. Since the responses in the dataset have been translated, we might misunderstand them, and this could make our sentiment analysis less accurate. Sometimes, tools like VADER might get the wrong idea, especially when a sentence sounds positive but actually means something negative. We also need to think about ethics. Things like privacy and whether the automated text analysis might have biases are important. So, this system should be more of a help than something that makes all the decisions on its own.

Reflection on feedback received and improvements

Based on the feedback received on Assignment 2, the report has been improved and refined in a number of areas.

- For Learning Goal 1, the pre-processing has been explained in greater detail. The steps are now explained more clearly and in a structured manner, as are our code and some additional explanation of why pre-processing is important for machine learning. This has made the process slightly more transparent and better justified.
- For Learning Goal 2, attention has been paid to the rationale behind the choices made. This applies to both the sentiment analysis (VADER) and the topic modelling, where it has been clearly explained why these methods were chosen and which parameters were used for them. We have also made it clearer why a combination of different analysis methods was used.
- Finally, in Learning Goal 3, the reasoning has been further developed, as the link between data analysis, insights and design choices has been made clearer. For instance, the solution description for each component explains how it arose from the results obtained from the analysis.

Overall, the report has therefore been improved by providing clearer steps, stronger justification and more detail behind the choices made, thereby better explaining the analysis and the design.

Machine Learning for Design



Assignment 3

Group: G 4

Members: Vlad Dobriakov, Josha Leemans, Karlijn Tangelder, Jop van der Post & Bing Zwikker

Introduction

In this project we developed a machine learning model that is able to classify the presence of heart disease using a combined dataset from the UCI repository (Switzerland and Hungary). During the data preprocessing we handled with missing values and removed values that were biologically impossible. This data cleaning process naturally transformed our target variable into binary values. We then trained and evaluated three models: a dummy classifier, K-Nearest Neighbours and a random forest classifier. To evaluate we used a 70/30 train-test split and a 3fold cross validation method. Our conclusion is that random forest demonstrated the best performance in recognizing complex patterns. Our analysis confirmed that features with high variance and medical relevance, such as maximum heart rate and ST depression, were the strongest predictors. Finally, to address class imbalance and further enhance model reliability, we proposed implementing SMOTE oversampling.

Data Preprocessing

Combining datasets

In the beginning we combined the two data sets together. Since both datasets originate from the same UCI Heart Disease dataset and share the same feature structure, they can be merged directly. Our primary motivation was to increase the number of samples, creating a more robust dataset for our machine learning model. However, since they come from different sources, we decided to add a dataset label before merging, to track the origin for each row and account for potential differences in feature distributions from the different hospitals. And at lastly we combined the 2 data sets in one data set named data without original index to prevent doubled index problems.

```
# Reading the data swiss =  
pd.read_csv('data/processed.switzerland.csv') hung =  
pd.read_csv('data/processed.hungarian.csv')  
  
# Add dataset label swiss["dataset"] =  
"swiss"  
hung["dataset"] = "hung"  
  
# Combine datasets  
data = pd.concat([swiss, hung], ignore_index=True)
```

After printing we saw that: each row represents a patient, and each column represents a medical feature such as age, blood pressure, cholesterol level, and heart-related measurements.

Based on the dataset description from the UCI repository, the features can be divided into numerical and categorical variables.

- **Numerical features include:** age, trestbps, chol, thalach, oldpeak, and ca
- **Categorical features include:** sex, cp, fbs, restecg, exang, slope, and thal

Dealing with missing values

A key observation is that several features contain a large number of missing values, especially ca and thal columns were almost entirely empty. Since imputing so much missing data would introduce heavy bias, we decided to drop these columns entirely. This was done by using the code part:

```
data = data.drop(columns=["ca", "thal"])
```

After that we dropped the other NaN values by using:

```
data = data.dropna()
```

After that we had 141 rows of data that are clean and ready to be used.

Structuring data

Also there could be considered to change some in the feature description. The cp and slope features are categorical but encoded as integers. These should ideally be transformed using one-hot encoding to avoid misleading the model with artificial ordering.

Now that we have a clean dataset, we used the following code to get the data statistics:

```
data.describe()
```

This function is to compute the distribution of the numerical data. This function automatically calculates key statistical measures such as the mean, standard deviation, and median. Providing a clear overview of the central tendencies and spread within our features.

After this all we can conduct the following:

	age	sex	cp	trestbps	chol	fb	restecg	thalach	exang	oldpeak	slope	target
count	141.000000	141.000000	141.000000	141.000000	141.000000	141.000000	141.000000	141.000000	141.000000	141.000000	141.000000	141.000000
mean	52.602837	0.808511	3.553191	136.978723	176.510638	0.092199	0.283688	123.056738	0.702128	1.259574	1.900709	1.191489
std	7.931744	0.394876	0.814384	19.078585	135.604710	0.290337	0.497512	22.853188	0.458953	0.951614	0.364593	0.955547
min	31.000000	0.000000	1.000000	92.000000	0.000000	0.000000	0.000000	60.000000	0.000000	-1.000000	1.000000	0.000000
25%	48.000000	1.000000	3.000000	120.000000	0.000000	0.000000	0.000000	105.000000	0.000000	0.800000	2.000000	1.000000
50%	53.000000	1.000000	4.000000	135.000000	219.000000	0.000000	0.000000	124.000000	1.000000	1.000000	2.000000	1.000000
75%	57.000000	1.000000	4.000000	150.000000	273.000000	0.000000	1.000000	140.000000	1.000000	2.000000	2.000000	1.000000
max	73.000000	1.000000	4.000000	200.000000	603.000000	1.000000	2.000000	184.000000	1.000000	5.000000	3.000000	4.000000

The dataset now contains 141 samples. The average age of patients is approximately 52.6 years, with a range from 31 to 73, indicating a middle-aged population.

Resting blood pressure (trestbps) has a mean of 136.98 mm Hg, with values ranging from 92 to 200, showing moderate variation across patients.

The maximum heart rate (thalach) has a mean of 123.06, with values between 60 and 184, indicating variability in patient fitness or heart condition. The feature oldpeak, which represents ST depression, has a mean of 1.26 and ranges from -1 to 5, where negative values may indicate measurement inconsistencies.

Removing invalid values

Several categorical features, such as sex, fbs, and exang, are binary (0 or 1), while others such as cp, restecg, and slope contain a limited set of discrete values, confirming that they should be treated as categorical variables rather than continuous ones.

Further inspection of the features revealed data entry errors that required preprocessing. For instance, the chol (cholesterol) feature contains values equal to 0, which is biologically impossible,

and oldpeak contains negative values (-1). To solve this, we will remove the rows containing these invalid entries with the following code:

```
data = data[(data['chol'] > 0) & (data['oldpeak'] >= 0)]
```

This code results in the following table:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	target
count	95.000000	95.000000	95.000000	95.000000	95.000000	95.000000	95.000000	95.000000	95.000000	95.000000	95.000000	95.000000
mean	50.631579	0.747368	3.463158	137.515789	261.978947	0.094737	0.231579	127.631579	0.736842	1.640000	1.905263	0.715789
std	6.860390	0.436827	0.884987	18.682603	68.979951	0.294405	0.424079	20.989893	0.442683	0.796455	0.328559	0.453430
min	31.000000	0.000000	1.000000	92.000000	117.000000	0.000000	0.000000	82.000000	0.000000	0.000000	1.000000	0.000000
25%	47.000000	0.500000	3.000000	124.500000	217.500000	0.000000	0.000000	113.500000	0.000000	1.000000	2.000000	0.000000
50%	52.000000	1.000000	4.000000	138.000000	260.000000	0.000000	0.000000	128.000000	1.000000	1.500000	2.000000	1.000000
75%	55.000000	1.000000	4.000000	150.000000	293.000000	0.000000	0.000000	142.000000	1.000000	2.000000	2.000000	1.000000
max	65.000000	1.000000	4.000000	200.000000	603.000000	1.000000	1.000000	184.000000	1.000000	5.000000	3.000000	1.000000

As visible in the new table the wrong data is successfully deleted. The minimum of chol is now 117 and oldpeak is 0, which means that the impossible values are deleted. The amount of rows has decreased from 141 to 95, which means we deleted $141 - 95 = 46$ rows.

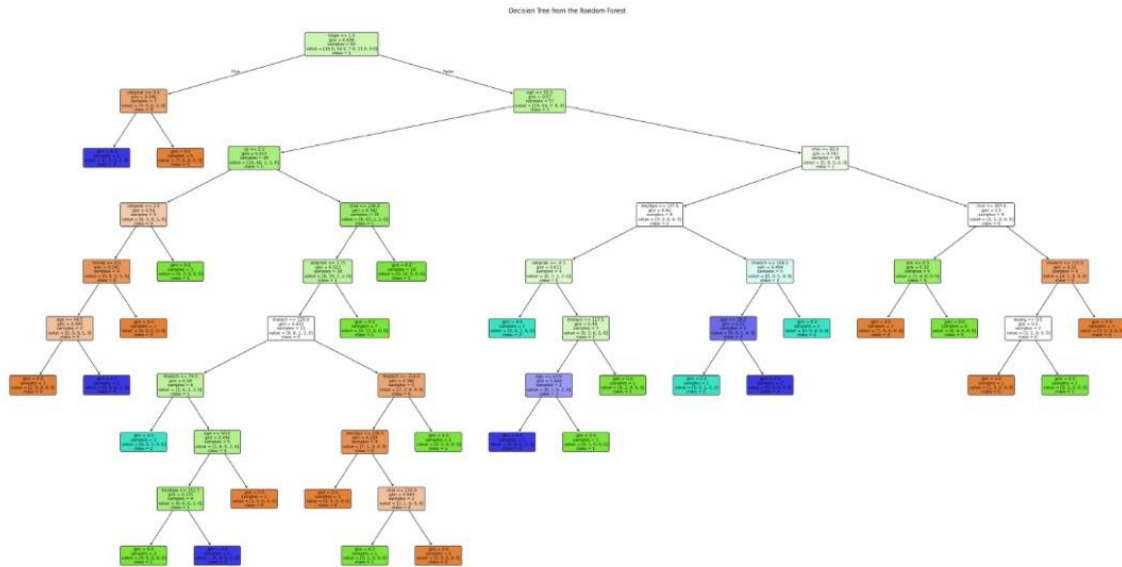
The target variable only has a minimum of 0 and a maximum of 1, which means we deleted the rows with target value 2, 3 or 4. This tells us that the people with a high level of heart disease were the people with the incorrect values of chol or oldpeak. We didn't make the target binary ourselves, but it's a side-effect of cleaning our data (with 0 = no heart disease and 1 = a present heart disease).

Model Training

Three different machine learning models were used to classify the presence of heart disease, namely: a Dummy Classifier, a K-Nearest Neighbours (KNN) classifier and a Random Forest classifier. These were chosen to represent an increasing level of complexity, ranging from a simple baseline to a model capable of identifying more complex patterns in the data. The Dummy Classifier was used as a reference point, as this model does not learn patterns from the dataset, but instead makes predictions based on a fixed rule. In this case, the “constant” strategy was chosen, whereby class 0 is predicted every time. The ultimate aim is to establish a baseline performance, so that it can be determined whether the other models are utilising any information from the data at all. The K-Nearest Neighbours classifier classifies the data points based on similarities. The distance to all points in the training data is calculated for all new data points, so that the k neighbours that are closest can then be selected. Here $k=4$ was chosen. The classification is ultimately determined by the class that occurs most frequently among these neighbours. A smaller value of k makes the model more susceptible to noise (overfitting), but a larger value can result in oversimplification (underfitting). Consequently, the choice of $k=4$ strikes a balance between the two extremes.

The Random Forest classifier is a more advanced model built on a collection of multiple decision trees. Each decision tree is trained on its own subset of the dataset and therefore makes an independent prediction. The final classification is determined by the class with the most predictions across all decision trees. In this study, we opted for a small number of trees, namely 4, so that the model can be kept computationally efficient and manageable, whilst still delivering better results than the simpler models. To better understand how decisions are made in a Random Forest, a visualisation was created for one of the decision trees. In such a tree, the data is split step by step based on features: for example, an initial split might occur on the “oldpeak” feature, where a check is made to see if the value exceeds a specific threshold. In this case, the data point is then sent to a branch where the probability of heart disease is higher. A subsequent split may then occur, such as on “thalach” (maximum heart rate). Through this series of decisions, a classification is ultimately made.

The visualisation of the decision trees shows that features such as “thalach” and “oldpeak” are used earlier in the decision-making process. This indicates that these variables hold considerable discriminatory power within the dataset. For example, higher values of “oldpeak” more frequently result in branches where the risk of heart disease is higher. This demonstrates that the model utilises relevant features to make predictions.

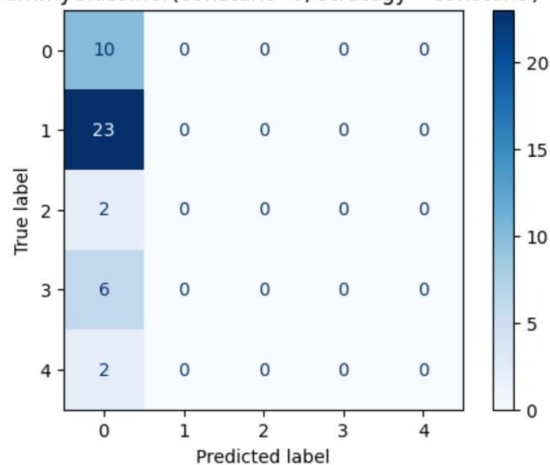


The model's learning processes were directly influenced by the pre-processing steps. Removing features with missing values, such as "thal" and "ca", reduced the complexity of the feature space. This reduces the likelihood of unreliable data causing the model to learn incorrect patterns, but may result in the loss of potentially relevant information. Unrealistic values, such as negative oldpeak and chol=0 values, have also been removed, making the dataset more consistent and, consequently, more reliable. This enables the models to better recognise relevant patterns

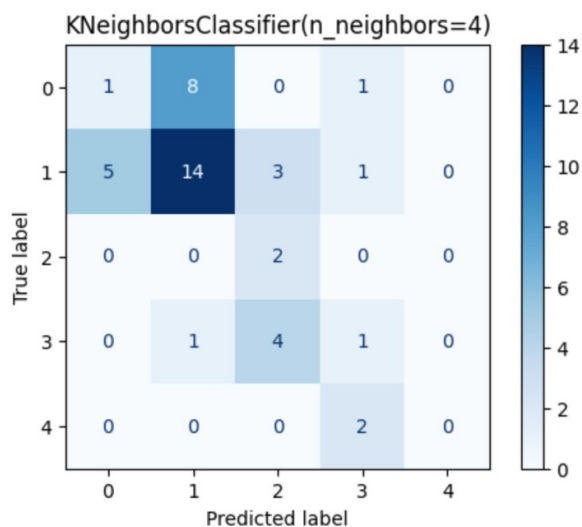
A consequence of the preprocessing is that the target variable has effectively become binary (0=no heart disease, 1=heart disease), as observations with higher severity values have been removed. This has simplified the classification task, as the models now only need to distinguish between the presence and absence of heart disease. In addition, the size of the dataset has decreased significantly, as the number of observations has fallen from 141 to 95, thereby increasing the risk of overfitting, particularly with more complex models such as the Random Forest. This highlights the importance of additional validation techniques.

Confusion matrices were created for each model to evaluate model performance. These matrices provide insight into correct and incorrect predictions per class, with the diagonal showing the correct predictions and the remaining values showing the misclassifications. The confusion matrix produced by the Dummy Classifier shows that all observations are classified as class 0. It can therefore be concluded that the model does not learn any patterns from the dataset and makes no distinction between the different classes.

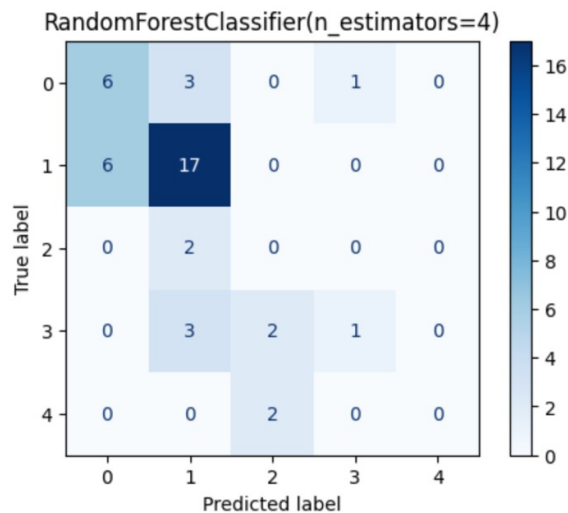
DummyClassifier(constant=0, strategy='constant')



The KNN model produces a confusion matrix that can predict multiple classes, but there are still a significant number of errors, which often occur between classes that are closely related. Observations from class 1 are regularly confused with those from class 0 or 2, which shows overlap in the feature space and are more difficult distinguished from one another based on the features.



In contrast, the Random Forest classifier shows a better distribution of correct predictions, with more values lying on the diagonal. This demonstrates that this model is better able to recognise underlying patterns in the dataset. Furthermore, the model continues to struggle with less common classes, as this is likely due to a poor distribution within the dataset.

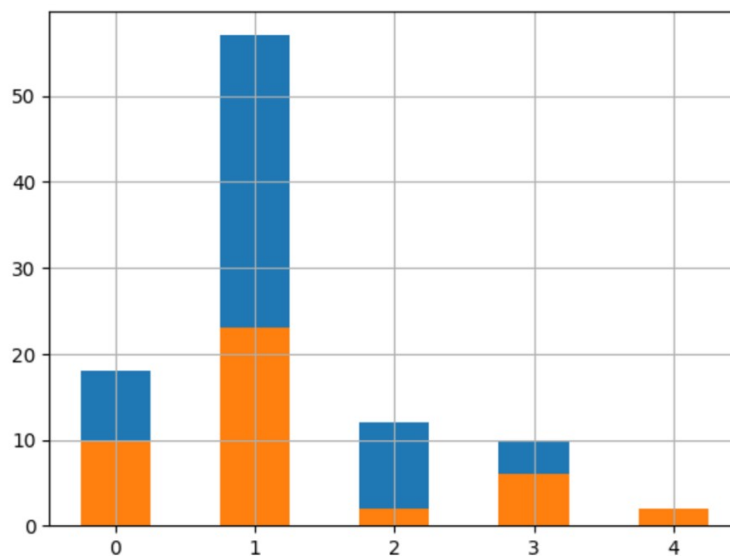


In addition to confusion matrices, other evaluation metrics are used, namely: accuracy, precision, recall and F1 score. However, whilst accuracy provides a general indication of the percentage of correct predictions, this metric is not particularly suitable for datasets with an uneven distribution. In this situation, recall is an important metric, as it indicates the extent to which the model correctly identifies patients with heart disease at all. False negatives can have serious consequences in a medical context, which is why the focus is on maximising recall. The F1-score, on the other hand, is used as a supplement to strike a balance between correctly identifying patients with the condition and minimising false predictions. The results show variation in the performance of classes. The lower recall for the less common class is less reliable due to the lack of observations. Furthermore, the dominant class performs better, demonstrating that bias is a consequence of “class imbalance”. This confirms that the use of multiple evaluation metrics is important for obtaining a complete picture of model performance.

	precision	recall	f1-score	support
0	0.17	0.10	0.12	10
1	0.61	0.61	0.61	23
2	0.22	1.00	0.36	2
3	0.20	0.17	0.18	6
4	0.00	0.00	0.00	2
accuracy			0.42	43
macro avg	0.24	0.38	0.26	43
weighted avg	0.40	0.42	0.40	43

The training and evaluation process has been standardised to enable reproducible and fair comparisons. All models were therefore trained under identical conditions and evaluated in the same way. The dataset has been split into 70% for training and 30% for testing, allowing the models to be tested on new data and providing a more realistic picture of how the models perform. The train-test split, as shown by the histogram, indicates that the class imbalance can be maintained in both the training and test sets, with one dominant class. This may therefore explain the superior performance of this class compared to the less common classes.

```
#### Save Histogram of Train/Test Label Counts ####
Histogram Saved
```



At the same time, 3-fold cross-validation was used, whereby the dataset was split into three parts and the model was trained and tested several times on different combinations of these splits. The final performance is determined by the average, resulting in a more robust and reliable evaluation. This approach results in a reproducible and partially automated experiment. It allows the models to be compared consistently, thereby providing a sound basis for further optimisation, such as testing different hyperparameters.

Discussion

The relationship between feature importance and data analysis

The feature importance derived from the Random Forest classifier largely corresponds to what we already observed in the data analysis from Task 1. This can be explained by the statistical properties of the features, such as variance, but also by the way a Random Forest model makes decisions.

The data analysis showed that certain numerical features, such as 'thalach', 'age', 'oldpeak' and 'trestbps', have a fairly wide range and high standard deviation. For example, 'thalach' has a standard deviation of 22.85 and a range from 60 to 184. This indicates that this feature varies greatly between patients, meaning it captures a lot of information to distinguish between the different levels of heart disease. The theory tells us that feature selection methods, such as VarianceThreshold in sklearn, often make features with low variance less useful, as they show little difference between the data points.

The Random Forest classifier determines feature importance based on how much a feature contributes to the splitting of data in the decision trees, by looking at the reduction in impurity, such as Gini impurity. Features that result in clear and consistent splits are given a higher importance score, as they reduce uncertainty in the predictions the most; consequently, it makes sense that features with a clear relationship and high variability with the target variable become more essential.

We see this reflected in features such as 'oldpeak' and 'thalach'. Due to the wide variation in "thalach", the model can draw boundaries between the different patient groups. In addition, "oldpeak" also has a clear medical significance, ST depression, making it even more relevant. This combination of substantive meaning and statistical variation ensures that these features contribute effectively to the model.

On the other hand, features such as "fbs", "restecg" and "sex" have low variance and often consist of values with limited variation or binary values. We saw in Task 1 that these features do not have much variation and are therefore less discriminatory. The theory also stated that features with lower variance are less informative. Given the nature of Random Forest, it therefore makes sense that these features are not used as frequently for splits and thus have a lower importance.

It is important to note, however, that feature importance does not depend solely on variance, as Random Forest can also detect complex and non-linear relationships between features. This indicates that a feature with lower variance can still sometimes be important if, in combination with other features, it provides meaningful information. In this case, however, the data analysis and the theoretical expectation point in the same direction, demonstrating that the model and the data are consistent with one another.

To summarise briefly: the feature importance from the Random Forest model aligns well with the statistical analysis from Task 1. Features that contain more variation and are closely related to heart disease in terms of content contribute most to the model's performance, whereas features with less variation have less influence. This demonstrates that the data analysis provides a solid basis for understanding how the model makes its decisions.

Enhancing the model's performance

One method of enhancing model performance involves addressing the dataset's imbalance. As was seen in the data exploration, the distribution of the target variable is skewed. Most of the patients belonged to one of the two groups (particularly, group one) with a few patients belonging to groups two and/or three. This degree of imbalance can lead to problems for the model because it will learn most of its patterns from the majority class and therefore, the model may not perform well when classifying instances from the minority classes. As such, the ability of the model to properly classify all groups may appear satisfactory even when the model is inaccurate at classifying those cases under-represented by their group.

To resolve this problem, we will implement resampling techniques, specifically oversampling, as the most appropriate approach for balancing the class distribution, primarily due to the relatively small size of our overall dataset. Oversampling increases the number of samples in the under-represented classes in order to create a more balanced distribution across the classes. This can be accomplished by creating duplicate copies of current samples or by creating new "synthetic" samples.

SMOTE is a common strategy used when attempting to oversample minority classes in an imbalanced dataset. Rather than creating copies of existing samples, SMOTE creates new synthetic samples by calculating the interpolation of a number of similar samples from the minor class. By training the model on these new samples, the model will learn a more generalised and meaningful representation of the behaviour of the minority class, rather than simply memorising duplicated data. As a result, the model will be able to more effectively distinguish between class labels.

Another method to address data imbalance is to remove samples from the majority class (undersampling). However, given that our dataset is relatively small, undersampling would result in the loss of a substantial amount of valuable data. In addition, removing a large number of samples from the majority class would negatively impact the model's validity and ability to generalise to new data. Therefore, oversampling will represent a better method of overcoming data imbalance in this case.

As a result of balancing the dataset, the model becomes less biased in favour of the majority class and is better able to identify patterns across all classes. This results in higher model performance, particularly when evaluating the minority classes based on recall and F1-score. Because accuracy alone is not an accurate representation of the model's performance when working with imbalanced datasets, recall and F1-score are the most appropriate measures of model performance when working with imbalanced datasets.

Ultimately, using oversampling techniques like SMOTE increases model reliability and fairness because it ensures that all classes are adequately represented in training data, helping the model to produce more accurate and fair predictions across heart disease severity levels.

Reflection on feedback received and improvements

Based on the feedback we received for Assignment 3, the 'Model Training' section has been improved to include a clearer explanation of the models used and an example of how decisions are made in a Random Forest. The choice of evaluation metrics has also been better justified, with the emphasis on recall and the F1 score. Finally, the training process has been described more clearly as a structured approach.